

Arunabha Majumder

Robotics Engineer | Mechanical Design, Motion Control, Actuators & Sensors, Hands-On Builds

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SOMETHING I'VE BUILT: A HYBRID-ACTUATED ROBOTIC SHOULDER JOINT

What it is

A robotic shoulder joint for an exoskeleton (PhD project VIXO, Aalborg University) that supports a person's arm against gravity. I designed the mechanics, built the prototypes, integrated the motor and sensors, implemented the control and built the test rigs. Project page: www.vixo.aau.dk

The problem

Carrying the full gravity load with a motor alone would need a big, heavy and power-hungry actuator, which is not practical for something worn on the body.

What I did

I paired a parallel spring, which carries the steady gravity load, with a small 6 Nm motor that only adds what the spring cannot. I designed the joint in SolidWorks, iterated through 3D-printed prototypes, and integrated the motor, encoder, force sensors, DAQ and embedded control. I then implemented an Assist-as-Needed controller and built test rigs to tune and check the system on the bench.

The result

The joint cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. In tests with 12 participants, the controller reduced muscle effort by more than 15%.

What I learned

Getting a machine to work reliably comes down to the details found on the bench and in testing, and I enjoy that part as much as the design.

Photos and more projects: arunabha22.github.io/arunabha-majumder